

SUMMER TERM 2024
DEPARTMENTALLY MANAGED ONLINE EXAMINATION
ECON0016: MACROECONOMIC THEORY & POLICY
TERM 2: INTERNATIONAL MACROECONOMICS

Assessment Component: 50% Remote Online Controlled Condition Examination

Page Limit: No page limit

All questions should be answered.

The True/False/Uncertain questions carry 10% of the total mark each, Problem I 30%, Problem II 10% and Problem III 30%.

True/False/Uncertain:

1. Consider an endowment small-open economy with lump-sum taxes. Suppose that the current account deficit is 5% of GDP and that half of the population (representing 50% of GDP) is borrowing-constrained. The other half is not subject to a borrowing constraint. The current account deficit in period 1 can be eliminated if the Government increases lump-sum taxes in period 1 by 10% of GDP, keeping Government expenditures constant in both periods. (no more than 500 words, less is fine, equations and graphs are allowed)
2. In a small open economy with production and perfect capital mobility (firms invest and households consume and save), a country with more impatient households (for example lower β if preferences are $\log C_1 + \beta \log C_2$) will consume more in period 1 and invest less in period 1. (no more than 500 words, less is fine, equations and graphs are allowed)
3. It is a fact that the real exchange rate (say *vis-à-vis* the US dollar) is decreasing with income per capita (rich countries are more expensive than poor countries). One explanation is that the relative price of nontraded goods with respect to traded goods is higher in poor countries, assuming that the Law Of One Price holds for traded goods. (no more than 500 words, less is fine, equations and graphs are allowed)

★

Problem I (Tradables, nontradables and the real exchange rate)

Consider a two-period, endowment, and small open economy. In period 1, households receive an endowment of 6 units of tradable goods and 9 units of nontradable goods. In period 2, households receive 13.2 units of tradables and 9 units of nontradables ($Q_1^T = 6$, $Q_2^T = 13.2$, and $Q_1^N = Q_2^N = 9$). Households start period 1 with no assets or liabilities ($B_0 = 0$). The country enjoys free access to world financial markets, where the prevailing interest rate is 10 percent ($r^* = 0.1$). Suppose that the household's preferences are defined over consumption of tradable and nontradable goods in periods 1 and 2, and are described by the following utility function:

$$U = \log C_1^T + \log C_1^N + \log C_2^T + \log C_2^N,$$

where C_t^T and C_t^N denote, respectively, consumption of tradables and nontradables in period $t = 1, 2$. Let p_1 and p_2 denote the relative prices of nontradables in terms of tradables in periods 1 and 2, respectively. We assume that bonds are expressed in units of the tradable good.

1. Write down the budget constraints of the household in periods 1 and 2.
2. Derive the household's intertemporal budget constraint. Assign this expression the number (1).
3. The household chooses consumption of tradables and nontradables in periods 1 and 2 to maximize its lifetime utility function subject to its intertemporal budget constraint. Derive the optimality conditions associated with this problem. To this end, begin by solving (1) for C_1^T and use the resulting expression to eliminate C_1^T from the lifetime utility function. Take the derivatives of the resulting lifetime utility with respect to C_1^N , C_2^T , and C_2^N and set them equal to zero. Assign the resulting three expressions the numbers (2), (3), and (4), respectively.
4. Write down the market-clearing conditions in the nontradable goods market in periods 1 and 2. Assign these expressions the numbers (5) and (6), respectively.
5. What is the value of r_1 under perfect capital mobility? Assign number (7) to that equation.
6. Combine expressions (1) to (6) to solve for C_1^T , C_2^T , C_1^N , C_2^N , p_1 , and p_2 to obtain the following

expressions:

$$\begin{aligned}
C_1^T &= \frac{1}{2} \left(Q_1^T + \frac{Q_2^T}{1+r^*} \right), \\
C_2^T &= \frac{1}{2} ((1+r^*)Q_1^T + Q_2^T), \\
C_1^N &= Q_1^N, \\
C_2^N &= Q_2^N, \\
p_1 &= \frac{1}{2Q_1^N} \left(Q_1^T + \frac{Q_2^T}{1+r^*} \right), \\
p_2 &= \frac{1}{2Q_2^N} ((1+r^*)Q_1^T + Q_2^T).
\end{aligned}$$

Explain why the relative price of nontradables changes over time.

7. Calculate the net foreign asset position of the economy at the end of period 1, B_1 .
8. Calculate the equilibrium levels of the current account balance in periods 1 and 2 (CA_1 and CA_2).
9. Assume that the domestic consumer price index in period $t = 1, 2$, denoted P_t , is defined by $P_t = \sqrt{P_t^T P_t^N}$, where P_t^T and P_t^N denote the nominal prices of tradables and nontradables in period $t = 1, 2$, respectively. Similarly, suppose that the foreign consumer price index is given by $P_t^* = \sqrt{P_t^{T*} P_t^{N*}}$, where the superscript $*$ denotes foreign variables. Foreign nominal prices are expressed in terms of foreign currency. Assume that PPP holds for tradable goods. Finally, suppose that the foreign relative price of nontradables in terms of tradables equals unity in both periods. Compute the real exchange rate in periods 1 and 2.

★

Problem II (Current Account Sustainability in a Three-Period Economy)

To address the questions below, derive step-by-step adaptation of the two-period economy we have studied in Chapter 2 of the course to a three-period horizon.

1. Can a three-period small-open endowment economy run a perpetual trade balance deficit?
2. Can it run a perpetual current account deficit?

★

Problem III (The Relative Price of Investment and the Current Account)

Consider a small two-period open-economy with production. It produces a tradable good that can be consumed or exported. The investment goods are not domestically produced and have to be imported. The price of investment goods in terms of consumption goods is equal to q_t . The economy is small in the sense that it cannot affect the price q_t . The production function of domestic firms is equal to $Y_t = A_t I_t^\alpha$, for $t = 1, 2$ and $0 < \alpha < 1$, where I_t denotes investment measured in units of investment goods. Preferences are $U = \log C_1 + \beta \log C_2$ where C_t is domestic consumption in period t and $0 < \beta < 1$. We denote r_t the interest rate and the world interest rate is r^* .

1. Find an expression for optimal I_1 (from profit maximisation) as a function of r_1 and q_1 . Indicate the sign of the derivative of this function with respect to r_1 and q_1 . Draw it in the plane (I_1, r_1) for given q_1 . How would an increase in q_1 shift this investment schedule?
2. Derive an expression for optimal Π_2 as a function of r_1 and q_1 .
3. Write down the maximization problem of the household under the budget constraints

$$C_1 + B_1^h = \Pi_1 + (1 + r_0)B_0^h \quad \text{and} \quad C_2 + B_2 = \Pi_2 + (1 + r_1)B_1^h,$$

that you have to collapse into an Intertemporal Budget Constraint. Using the first order condition and the IBC, find an expression for optimal C_1 as a function of r_1 and q_1 . Indicate the sign of the derivative of this function with respect to r_1 and q_1 . Draw it in the plane (C_1, r_1) for given q_1 . How would an increase in q_1 shift this consumption schedule?

4. Find an expression for national savings S_1 as a function of r_1 and q_1 . Indicate the sign of the derivative of this function with respect to r_1 and q_1 . Draw it in the plane (S_1, r_1) for given q_1 . How would an increase in q_1 shift this saving schedule?
5. Find an expression for the current account in period 1 CA_1 as a function of r_1 and q_1 . Indicate the sign of the derivative of this function with respect to r_1 and q_1 . Draw it in the plane (CA_1, r_1) for given q_1 . How would an increase in q_1 shift this current account schedule?
6. Suppose that the economy is small and open to international trade in goods and assets and that free capital mobility prevails. What is the effect of an increase in q_1 on the equilibrium levels of saving, investment, and the current account? Answer the question graphically.
7. How does the answer to the previous question change if the economy is assumed to be closed to trade in financial assets but open to trade in goods?